

Improving BPMN Model Generation from Texts through a Prompt-Guided Fusion of NLP and LLM Approaches

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Abstract

The automatic generation of BPMN models from textual descriptions remains a major challenge, particularly when dealing with long and complex texts. Traditional natural language processing (NLP) approaches struggle with linguistic disambiguation, often resulting in incorrect BPMN models characterized by overloaded or missing relevant process elements. On the other hand, large language models (LLMs) such as GPT-4, while effective at capturing semantic nuances, tend to omit key elements required for complete modeling. The objective of this study is to improve the quality of automatically generated BPMN models from textual descriptions by overcoming the respective limitations of traditional approaches and GPT-4. To this end, we propose a prompt-guided fusion method that combines a literature-based NLP approach with the reasoning capabilities of GPT-4. This fusion helps reduce the overload of activities and exclusive gateways, particularly for the NLP approach, while partially correcting errors related to the generation of parallel gateways for both approaches.

CCS Concepts

• Computing methodologies; • Natural language processing;;
• Knowledge representation and reasoning;; • Information
systems; • Business process modeling;

Keywords

NLP, LLM, BPMN models, textual descriptions, fusion method

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1 Introduction

Business process modeling plays a central role in improving organizational efficiency by enabling better understanding, execution, and automation of operations. However, the automatic generation of these models from textual descriptions remains a major research challenge, primarily due to the complexity and ambiguity of natural language.

In recent years, several studies have investigated the potential of natural language processing (NLP) to convert textual descriptions into BPMN diagrams. For instance, Ferreira et al. [1] extracted process model elements from text without generating a complete BPMN diagram. Similarly, Epure et al. [2] leveraged semi-structured textual input but did not produce a BPMN diagram.

The approach by Friedrich et al. [3] stands out by enabling the automatic generation of complete BPMN diagrams from unstructured textual descriptions. It combines natural language processing techniques, an intermediate semantic model called the *World Model*, and a multi-step transformation procedure that results in a readable and standards-compliant BPMN representation. Nevertheless, their evaluation shows that this transformation approach tends to produce overly detailed models, with a higher number of nodes and edges than necessary.

With the emergence of large language models (LLMs), several recent studies have explored their significance in this field, while others have highlighted certain limitations. For example, Speck et al. [14] used ChatGPT to transform legal rules into formal BPMN models, often identifying tasks correctly, but sometimes omitting important activities and fail to generate the final exclusive gateways. Daclin et al. [13] studied also used ChatGPT to convert textual descriptions into process models, finding that direct input causes modeling errors, including over- and under-modeling of activities.

In this article, we propose a fusion method based on a prompting technique that combines the textual BPMN models generated by [3] and those produced by GPT-4, to create a final BPMN model aimed at reducing the errors encountered by each of these approaches and aligning more closely with the manual models.

The article is structured as follows: Section 2 presents a review of related work, Section 3 outlines the preliminary analysis and the proposed method, Section 4 is devoted to the obtained results, and Section 5 concludes the article.

2 Related Work

The analysis of related work is structured around two categories of approaches for extracting BPMN models from textual descriptions. The first concerns NLP-based approaches, while the second focuses on LLM-based approaches. The specific challenges associated with each of these approaches are presented.

NLP-based approaches

Natural language processing (NLP) techniques rely on syntactic analysis, lexical databases, and rule-based extraction to identify process elements.

The fundamental work of Friedrich et al. [3], which is considered the state of the art according to recent studies [5]. Our study builds upon the process models produced by this approach. While their work focuses on the automatic generation of a BPMN model from a textual description, our study aims to improve these models to bring them closer to manual reference models. The authors combine syntactic features with information from a lexical database to identify patterns at both the sentence and document levels, with the goal of creating BPMN models [6]. However, this approach has significant limitations observed in several studies: it does not handle the complexity of certain natural language structures, particularly complex or compound sentences [7], and often generates superfluous activities [15]. Gonçalves et al. [8] introduce storytelling mining, which generates BPMN models from user stories. The approach combines text mining, syntactic rules, and lexical triggers to extract actors, activities, and gateways, before building the model. However, it remains limited, as it does not handle complex syntactic structures and has only been tested on two case studies. Honkisz et al. [9] incorporate a semantic analysis that filters out irrelevant Subject-Verb-Object (SVO) structures to retain only valid activity names. The results are first presented as spreadsheets before being converted into BPMN models. However, this method has not been tested on inputs containing complex sentences. Additionally, gateway extraction remains limited, and another issue concerns the extraction of invalid activities from descriptions.

LLM-based approaches

The recent emergence of pre-trained Large Language Models (LLMs), endowed with advanced reasoning capabilities, opens new opportunities for the automatic generation of BPMN (Business Process Model and Notation) models from textual descriptions. Among the most notable works:

Kourani et al. [10] utilize an intermediate format (POWL) to ensure the consistency and executability of the generated models; Köpke and Safan [11] present BPMN-Chatbot, a conversational tool designed to produce accurate and accessible models even for novice users; and Nour Eldin et al. [12] propose Nala2BPMN, which combines LLMs and deterministic algorithms to handle complex textual descriptions.

While these works highlight the importance of LLMs in BPMN model generation, they do not address issues related to process elements that LLMs may generate incorrectly. Some studies, however, have highlighted these limitations, including:

Daclin et al. [13] investigated the use of LLMs, such as ChatGPT, to automatically transform textual process descriptions into BPMN models and highlighted that direct use of LLMs frequently leads to over-modeling issues, including the addition of activities not

present in the text, as well as under-modeling, with the omission of key elements, and that direct generation remains mainly reliable for simple processes. Speck et al. [14] used ChatGPT to analyze and transform legal rules written in natural language into process models. They show that ChatGPT often succeeds in identifying text segments as tasks, but some important activities are omitted, even though they appear in the manual model. Moreover, the final exclusive gateways are not always generated correctly. Grohs et al. [15] leveraged GPT-4 using structured prompts for process model generation. However, they observed that the models produced by GPT-4 generally contain an insufficient number of tasks. Brissard et al. [16] evaluated different process model representations (PMRs) generated by LLMs. Their study shows that LLMs tend to under-generate exclusive and parallel gateways. This systematic under-detection leads to simplified models that do not faithfully reflect the expected control flow complexity.

As shown by previous studies based on NLP and LLM approaches, the generated BPMN models often contain superfluous or missing activities and under-generate gateways. These limitations motivate the development of our approach, which aims to combine NLP and LLM techniques to produce BPMN models that are both more accurate and more complete.

3 Preliminary Analysis and Proposed Method

Based on the limitations highlighted in related work concerning NLP and LLM, which we have taken into account in order to overcome them, we conducted an preliminary analysis prior to presenting the proposed method. Specifically, we evaluated the models generated by the NLP approach, considered state-of-the-art, on six process descriptions from [4] (1.1–1.4, 2.1, and 2.2), selected to include both short and simple descriptions as well as longer and more complex ones. We also evaluated the models produced by the LLM using a structured prompt, comparing them with the manual models.

The objective of this analysis is to highlight the discrepancies observed between the models, in order to justify the proposed method. This approach aims to address the main limitations identified in the literature as well as those revealed by our own analyses.

3.1 Preliminary Analysis

To analyze the different models, we first compared the models generated by the NLP approach of Friedrich with the manual models. Then, we compared the models produced by the LLM with these same manual models.

The comparison was performed by calculating, for each model, the number of the following process elements: activities, XOR gateways, and AND gateways.

3.1.1 Comparison of NLP and Manual Models. Table 1 presents the numbers of activities and gateways generated by the NLP model compared to the manual model (GS).

This table highlights three key limitations of the NLP model:

- **Overgeneration of activities** across all texts compared to the GS (e.g., 56 vs. 26 in text 2.1).
- **Excessive XOR gateways detection** in very long texts (e.g., 17 vs. 6 in text 2.1).

Table 1: Activities and Gateways in NLP vs. GS Models

Text	Type	Activities		XOR Gateways		AND Gateways	
		PM	GS	PM	GS	PM	GS
1.1	Long	18	9	6	4	2	2
1.2	Court	10	8	3	3	0	0
1.3	Long	21	12	3	4	4	3
1.4	Long	19	7	2	2	2	3
2.1	Very long	56	26	17	6	0	2
2.2	Very long	46	20	9	6	0	2

Abbreviations: NLP = Model by Friedrich et al.; GS = Gold Standard.

Table 2: Activities and Gateways in LLM vs. GS Models

Text	Type	Activities		XOR Gateways		AND Gateways	
		LLM	GS	LLM	GS	LLM	GS
1.1	Long	9	9	2	4	2	2
1.2	Court	8	8	2	3	1	0
1.3	Long	7	12	1	4	2	3
1.4	Long	10	7	1	2	0	3
2.1	Very long	33	26	4	6	1	2
2.2	Very long	21	20	5	6	0	2

Abbreviations: LLM = Large Language Model (GPT-4); GS = Gold Standard.

- **Missed AND gateways** in very long texts, where parallelism is expected (e.g., 0 vs. 2 in texts 2.1 and 2.2).

3.1.2 Generation of LLM Models. To generate models using a Large Language Model (LLM), we employed GPT-4 as the specific implementation in our study. For consistency, we refer to it as the LLM throughout the rest of this paper. We reused a prompt taken from the work of [15], which was limited to extracting flow and actor-executor relationships.

This prompt was selected over others because it includes the following essential BPMN elements: tasks, XOR/AND gateways, arrows, and actors associated with activities. It guides the LLM in producing an intermediate textual representation of the BPMN process.

3.1.3 Comparison of LLM and Manual Models. Once the models produced by the LLM were generated, we compared them with the manual model to identify their limitations. Table 2 presents the number of activities and gateways produced by the LLM compared to the manual model.

As shown in Table 2, the models generated by the LLM are generally better than those produced by the NLP model. However, some limitations remain:

- **Inconsistent activity detection:** the LLM tends to under-produce activities for long texts (e.g., 7 vs. 12 in text 1.3) but can overproduce activities for very long texts (e.g., 33 vs. 26 in text 2.1).
- **Under-identification of XOR gateways:** the LLM generates fewer exclusive gateways compared to the GS (e.g., 2 vs. 4 in text 1.1).

- **Insufficient detection of AND gateways:** the LLM detects fewer parallel gateways than the GS, and sometimes misses them, similar to the NLP model.

3.2 Proposed Method

This section describes the proposed fusion method, which combines the two textual BPMN models produced by Friedrich’s approach and the LLM, while relying on the textual description as a validation source. The objective of this method is to overcome the limitations previously observed in both models as well as those presented in the works discussed in Section 2.

First, we manually transformed the process diagrams from [4] into a textual format consistent with that produced by the LLM, making them usable in the fusion process. Once the two textual representations were obtained (Friedrich’s textual model and the LLM’s textual model), we used a prompt to perform an intelligent fusion of the two models, following a structured three-step approach :

Step 1 : Retain Common Elements

The first step consists of identifying elements present in both automatically generated models (LLM and Friedrich), including activities, events, sequence flows, or actors. These elements are considered reliable and essential, as their redundant presence suggests a convergence of interpretation between two different approaches. They are therefore automatically retained in the fused model without modification. This step minimizes the risk of omitting important elements. Even though our comparison in the analysis section does not focus on all the elements, since they are generated in both models, it is preferable to improve them.

Table 3: Activities and Gateways in PM vs. GS Models

Text	Type	Activities		XOR Gateways		AND Gateways	
		PM	GS	PM	GS	PM	GS
1.1	Long	10	9	3	4	2	2
1.2	Court	9	8	2	3	1	0
1.3	Long	11	12	1	4	2	3
1.4	Long	12	7	1	2	2	3
2.1	Very long	39	26	5	6	1	2
2.2	Very long	22	20	5	6	1	2

Abbreviations: PM = Proposed method; GS = Gold Standard.

Step 2 : Include Unique Elements Only if Justified by the Text

The second step examines elements that appear in only one of the models. These elements can be activities, actors, or other process components. Their inclusion in the merged model is conditional: they are retained only if they are clearly justified by the textual description of the process. This means checking that adding the element is consistent with what is described, to avoid introducing unnecessary or incorrect parts. This step allows the model to be completed by adding elements that may have been missed by the other model. For example, Friedrich tends to generate too many activities, often of little use, whereas the LLM may omit some (e.g., Text 1.3). The goal is to leverage each approach's strengths without unnecessarily cluttering the diagram.

Step 3 : Retain or Add Gateways (XOR/AND) Only if the Text Requires

This final step concerns decision (XOR) and parallelism (AND) gateways. Instead of keeping all gateways from both models, a gateway is retained or added only if the text indicates an exclusive choice logic (e.g., "if... then... else...") for XOR, or a parallel activity (e.g., "at the same time," "while...") for AND. This approach helps avoid two issues: an excess of unnecessary gateways, which fragment the model and make it complex (a common problem with Friedrich), and missing gateways, which render the model incomplete or incorrect (as can sometimes occur with the LLM). Additionally, a gateway that is absent in both models but mentioned in the text can be added to ensure the final model is coherent and complete.

These three steps address the three limitations observed during our analysis. Finally, after generating the final model with the LLM, we automated its conversion into a process diagram using Python ProcessPiper, which generates a PNG image from a simplified textual syntax.

4 Result

Table 3 shows that our fused method (PM) improves the quality of BPMN models compared to the standalone NLP and LLM approaches:

- **Activities:** The number of generated activities is closer to the manual model (GS), avoiding both the overgeneration from the NLP model and the omissions from the LLM (e.g., 11 vs. 12 in text 1.3).

- **XOR Gateways:** The fused model limits the excess found in the NLP approach and compensates for the under-identification by the LLM (e.g., 3 vs. 4 in text 1.1).

- **AND Gateways:** Although partial, the improvement is notable for complex texts where both base approaches failed (e.g., 1 vs. 2 in text 2.2).

Overall, the final model is closer to the manual model, effectively leveraging the strengths of the NLP and LLM approaches. It corrects the flow order, avoids many irrelevant activities and exclusive gateways, and sometimes adds elements missing in both models. Although it does not exactly match the manual model, it significantly improves the quality of both models.

5 Conclusion

In this article, we proposed a fusion method combining an NLP approach from the literature with a LLM. This synergy aims to overcome the limitations inherent to each approach when used in isolation, in order to improve the quality of BPMN models automatically generated from textual descriptions.

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